

Krishna Ramamoorthy

Curriculum Vitae

Santa Clara University, Computer Science & Engineering, 500 El Camino Real, Santa Clara, CA 95053

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EDUCATION

Doctor of Philosophy

Aug 2018 – Jun 2023

Computational Science

University of California, Irvine & San Diego State University

Master of Science

Aug 2014 – May 2016

Electrical Engineering

California State University, Northridge

Bachelor of Technology

Jun 2010 – May 2014

Electronics and Communications Engineering

Amrita Vishwa Vidyapeetham, India

ACADEMIC APPOINTMENTS

Assistant Professor (Tenure-Track)

Sept 2024 – Present

Department of Computer Science and Engineering

Santa Clara University

Lecturer

May 2023 – Jun 2024

Department of Computer Science

San Diego State University

Teaching Associate (AY)

Aug 2021 – May 2023

Department of Computer Science

San Diego State University

PROFESSIONAL EXPERIENCE

System Architect

Feb 2017 – Aug 2018

Kaiser Permanente IT, Pasadena, CA

Test Engineer Intern

Jan 2016 – May 2016

Aruba, a Hewlett Packard Enterprise Company, Sunnyvale, CA

GRANTS

Skills-Based Boot Camps and Workshops Grant – \$5,000 High Performance Computing (HPC) Center, Santa Clara University	<i>2026</i>
Grow Grant – \$1,000 Santa Clara University	<i>2025</i>
2FURS (Faculty-mentored Undergraduate Research) Grant – \$1,000 Santa Clara University	<i>2025</i>
Catalytic Project-Based Grant – \$5000 High Performance Computing (HPC) Center, Santa Clara University	<i>2025</i>
University Research Grant – \$4,976.44 Santa Clara University	<i>2025</i>
2FURS (Faculty-mentored Undergraduate Research) Grant – \$1,000 Santa Clara University	<i>2024</i>
Student Success Fee (SSF) Grant – \$5,000 San Diego State University	<i>2024</i>
Student Travel Grant – IEEE WCNC - \$750 IEEE Wireless Communications and Networking Conference (WCNC), Austin, TX, USA	<i>2022</i>
Student Success Fee (SSF) Grant – \$7,000 San Diego State University	<i>2021</i>
UCI Associated Graduate Students Grant – \$600 University of California, Irvine	<i>2020</i>
Amrita TBI Seed Grant – INR 25,000 (\$400 approx.) Technology Business Incubator (TBI), Amrita Vishwa Vidyapeetham, India	<i>2014</i>

AWARDS AND HONORS

Career Influencer Award Career Center, Santa Clara University	<i>2025</i>
Advancing AANHPI Educational Equity Scholarship - \$500 Bill & Melinda Gates Foundation (BMGF)	<i>2022</i>
Deborah M. Dexter Endowed Scholarship – \$888.50 San Diego State University — Awarded for academic excellence	<i>2020</i>
ACM Student Research Competition – Finalist 34th IEEE/ACM International Conference on Automated Software Engineering	<i>2019</i>
Natural Science, Inc. Best Research Award – \$250 ACSESS for Industry Conference	<i>2019</i>
Scholastic Achievement Award California State University, Northridge	<i>2016</i>

INVITED TALKS

Whats next in wireless: From Wi-Fi to 6G Alumini Meeting, Santa Clara University	May 2026
Intelligent Traffic Distribution in Wi-Fi 7 Multi Link Network Advisory Board Meeting, Santa Clara University	May 2025

Pricing Strategies to Improve User Experience in Future 5G Communications
Colloquium at Computational Science Research Center, San Diego State University

Apr 2023

INVITED PANELS

Higher Ed in STEM

Asian Pacific Islander Desi American Center
San Diego State University

Oct 2023

TEACHING EXPERIENCE

Computer Networks (CSEN 146)

Santa Clara University
Developed original course content.

Spring 2026

Advanced Programming (CSEN 11)

Santa Clara University
Developed original course content and redesigned the labs. Developed a new set of labs to align classwork with the lab.

Winter 2026

Design and Implementation of Programming Languages (CSEN 171)

Santa Clara University
Developed original course content. Developed 4 new practicums to provide students with hands-on programming experience in various programming paradigms.

Fall 2025, Fall 2026

Object-Oriented Programming and Advanced Data Structures (CSEN 79)

Santa Clara University
Developed original course content with a strong focus on in-class problem solving and coding. Introduced a term project component to give students hands-on experience applying the material.

Winter 2025

Computer Networks (CSEN 233)

Santa Clara University
Developed new instructional material to emphasize practical application of network protocols.

Fall 2024

Operating Systems (CS 480)

San Diego State University
Converted the course into a fully project-based learning experience. In this accelerated 6-week format, students worked in groups to implement a new project each week based on the lecture topics.

Summer 2023, Summer 2024

Computer Organization (CS 240)

San Diego State University
Redesigned the course and lab structure to reflect modern industry requirements. Built a lightweight autograder to help students get instant feedback on their codes — a feature that was later adopted by other instructors.

Spring 2022, Fall 2022, Spring 2023, Fall 2023, Spring 2024

Advanced Programming Languages (CS 420)

San Diego State University
Introduced Haskell into the curriculum for the first time. Designed new lab modules to explore type systems and functional programming.

Fall 2023, Spring 2024

Intermediate Programming (CS 160)

San Diego State University
Created in-class exercises for students transitioning from beginner to intermediate JAVA programming.

Fall 2021

PUBLICATIONS

Student co-authors I supervised are denoted with an asterisk (*).

1. D. Chui*, W. Wang, **K.M.K. Ramamoorthy**, “Semantic-Aware Learning-Based NOMA User Grouping for Programmable 6G RANs,” in *Proc. IEEE Wireless Communications and Networking Conference (WCNC)*, Apr. 2026.

2. A. Rajpurohit*, M. Kelley*, W. Wang, **K.M.K. Ramamoorthy**, "BALANCE: Bitrate-Adaptive Limit-Aware Netcast Content Enhancement Utilizing QUBO and Quantum Annealing," in *Proc. IEEE Wireless Communications and Networking Conference (WCNC)*, Mar. 2025.
3. G. Esitashvili, **K.M.K. Ramamoorthy**, W. Wang, Y. Zhao, "Study of Reconfigurable Intelligent Surface Deployment for Non-Orthogonal Multiple Access Wireless Communication Networks," in *Proc. of the International Symposium on Intelligent Computing and Networking*, Aug. 2024.
4. T. Kocher*, S. Braude* and **K.M.K. Ramamoorthy**, "Quantum-Accelerated Nash Equilibrium Search for Optimal Relay Selection in Wireless Networks", in *Proc. 2024 Intermountain Engineering, Technology and Computing (IETC)*, May 2024.
5. **K.M.K. Ramamoorthy**, W. Wang, K. Sohraby, Y. Zhao, "Proof-of-QoE NOMA Token: A Crypto Rewarding Concept To Incentivize Local Relay In Non-Orthogonal Multiple Access Wireless Networks," *International Conference on Computing, Networking and Communications (ICNC) Workshop on Computing, Networking and Communications (CNC)*, Feb. 2024.
6. Y. Song, **K.M.K. Ramamoorthy**, W. Wang and K. Sohraby, "A Use-It-Or-Lose-It Economic VCG Auction Approach For NOMA Wireless Relay Networks," *2023 IEEE International Conference on Omni-layer Intelligent Systems (COINS)*, Berlin, Germany, 2023, pp. 1-6.
7. **K.M.K. Ramamoorthy**, W. Wang and K. Sohraby, "Incentivize Non-Orthogonal Multiple Access In Wireless Multimedia Communications," *2023 IEEE Wireless Communications and Networking Conference (WCNC)*, Glasgow, United Kingdom, 2023, pp. 1-6
8. **K.M.K. Ramamoorthy**, W. Wang and K. Sohraby, "Orthogonality-Centric Pricing for Wireless Multimedia Multiple Access Networks," *ICC 2022 - IEEE International Conference on Communications*, Seoul, Korea, Republic of, 2022, pp. 2513-2518
9. E. Ballesteros, **K.M.K. Ramamoorthy** and W. Wang, "Exploring AV1 Encoder Potentials for Priority-Driven Wireless Multimedia Services," *2022 Intermountain Engineering, Technology and Computing (IETC)*, Orem, UT, USA, 2022, pp. 1-6.
10. **K.M.K. Ramamoorthy** and W. Wang, "Human Cognition Aware QoE For NOMA Pricing: A Prospect-Theoretic Augmentation To Non-Orthogonal Wireless Multiple Access," *2022 Intermountain Engineering, Technology and Computing (IETC)*, Orem, UT, USA, 2022, pp. 1-5.
11. **K.M.K. Ramamoorthy**, W. Wang and K. Sohraby, "NOMA Resource Block As A Commodity Box: Content-Centric QoE-Price Interplay In Wireless Multimedia Communications," *2022 IEEE Wireless Communications and Networking Conference (WCNC)*, Austin, TX, USA, 2022, pp. 2673-2678.
12. **K.M.K. Ramamoorthy** and W. Wang, "A QoE-Driven Pricing Scheme for Inter-Vehicular Communications With Four-Stage Stackelberg Game," in *IEEE Transactions on Vehicular Technology*, vol. 71, no. 3, pp. 3121-3130, March 2022.
13. **K.M.K. Ramamoorthy** and S. Mirzaei, "Design and Implementation of IoT based Cloud enabled Wireless Biometric Monitoring Device," *2021 IEEE 12th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, Vancouver, BC, Canada, 2021, pp. 0530-0533.
14. **K.M.K. Ramamoorthy**, W. Wang and K. Sohraby, "NOMAP: A Pricing Scheme for NOMA Resource Block Selection and Power Allocation in Wireless Communications," *2021 IEEE International Symposium on Local and Metropolitan Area Networks (LANMAN)*, Boston, MA, USA, 2021, pp. 1-6.
15. **K.M.K. Ramamoorthy** and W. Wang, "Profit-Driven Cache Delegation: A Game-Theoretic Wireless Multimedia Offloading Solution," *ICC 2021 - IEEE International Conference on Communications*, Montreal, QC, Canada, 2021, pp. 1-6.
16. **K.M.K. Ramamoorthy** and W. Wang, "Prospect Theoretic Pricing For QoE Modeling In Wireless Multimedia Networking," *2020 Intermountain Engineering, Technology and Computing (IETC)*, Orem, UT, USA, 2020, pp. 1-6.
17. **K.M.K. Ramamoorthy** and W. Wang, "QoE-Sensitive Economic Pricing Model for Wireless Multimedia Communications Using Stackelberg Game," *2019 IEEE Global Communications Conference (GLOBECOM)*, Waikoloa, HI, USA, 2019, pp. 1-6.
18. **K.M.K. Ramamoorthy**, "User Preference Aware Multimedia Pricing Model using Game Theory and Prospect Theory for Wireless Communications," *2019 34th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, San Diego, CA, USA, 2019, pp. 1265-1267.
19. **Ramamoorthy, K.M.K.**, Wang, W., Sohraby, K. (2019). "Stackelberg Game-Theoretic Spectrum Allocation for QoE-Centric Wireless Multimedia Communications". In: Zhang, T., Wei, J., Zhang, L.J. (eds) *Edge Computing – EDGE 2019*. *EDGE 2019. Lecture Notes in Computer Science*, vol 11520. Springer, Cham.

20. Vallur, B.P., **Ramamoorthy, K.M.K.**, Mirzaei, S., Mirzai, S. (2019). "Cerebral Blood Flow Monitoring Using IoT Enabled Cloud Computing for mHealth Applications". In: Arai, K., Kapoor, S., Bhatia, R. (eds) *Advances in Information and Communication Networks. FICC 2018. Advances in Intelligent Systems and Computing*, vol 887. Springer, Cham.
21. Littlewood, P., Mirzaei, S., **Ramamoorthy, K.M.K.** (2018). "Reconfigurable IP-Based Spectral Interference Canceller". In: Voros, N., Huebner, M., Keramidas, G., Goehring, D., Antonopoulos, C., Diniz, P. (eds) *Applied Reconfigurable Computing. Architectures, Tools, & Applications. ARC 2018. Lecture Notes in Computer Science*, vol 10824. Springer, Cham.
22. P. Littlewood, **K.M.K. Ramamoorthy** and S. Mirzaei, "Modeling of digital baseband interference canceler using Hilbert and Fourier Transforms," *2017 IEEE 13th International Colloquium on Signal Processing & its Applications (CSPA)*, Penang, Malaysia, 2017, pp. 123-128.
23. A. Chandramohan, **K.M.K. Ramamoorthy**, G. Sowmya, P.A. Surya Prasad, V. Vijay Krishna, and K.P. Peeyush, "Cost effective object recognition and sorting robot using embedded image processing techniques," in *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*, Apr. 2014.

SERVICE TO PROFESSION

Technical Program Committee - Member

IEEE 51st Conference on Local Computer Networks (LCN) - 2026
 IEEE Global Communications Conference (Globecom) - Track 1: Communication & Information System Security - 2026
 IEEE International Conference on Communications and Computer Networks (ICCCN) - Track 5: Mobile/Pervasive Communications, Networking and Sensing - 2026
 IEEE Wireless Communications and Networking Conference (INFOCOM) - Demo - 2025
 IEEE International Conference on Computing, Networking and Communications (ICNC) - Track 5: Multimedia Computing and Communications (MCC) - 2024
 IEEE Intermountain Engineering, Technology and Computing Conference (IETC) - 2024

Technical Reviewer

IEEE Wireless Communications and Networking Conference (WCNC): 2023, 2024, 2025
 IEEE Global Communications Conference (Globecom): 2019
 KDD 2025 Undergraduate and Masters Consortium - 2025

Journal Reviewer

IEEE Wireless Communications Magazine — 2024, 2025
 Results in Optics — 2023
 International Journal of Electrical, Electronics and Computer Systems (IJECS) — 2021

Judge

Student Research Symposium, San Diego State University — 2023

Faculty Advisor

Association for Computing Machinery (ACM) Student Chapter, Santa Clara University — 2026–Present
 Aztec Quantum Computing Club, San Diego State University — 2023–2024
 Girls Who Code Chapter at San Diego State University — 2023–2024

STUDENT SUPERVISION

Student Mentees, Santa Clara University

Graduate Students

• Tom Ngo	2026–Present
• Khushi Salaliya	2025–Present
• Zachary Common	2025–Present
• Brian Trinh	2024–Present
• Mrudhula Lokesh	2024–Present
• Samarth Kulkarni	2024–2025
• Avani Vaidya	2024

Undergraduate Students

• Guangyi Liu	2026–Present
• Collin Fiske	2025–Present
• Ariana Sun	2025–Present

• Andy Li	2025–Present
• Stephanie Campos	2025–Present
• Sean Lai	2025–Present
• Derek Chui	2024–Present

Student Mentees, San Diego State University

Undergraduate Students

• Michael Kelley	2023–2025
• Animesh Rajpurohit	2023–2025
• Riley Thompson	2023–2025
• Ashley Olson	2024–2025
• Tanner Kocher	2023–2024
• Samuel Braude	2023–2024

PROFESSIONAL MEMBERSHIPS

Member, Sigma Xi – The Scientific Research Honor Society	2026–Present
Member, Institute of Electrical and Electronics Engineers (IEEE)	2021–Present
Member, Institute of Electrical and Electronics Engineers (IEEE)	2021–Present

REFERENCES

Dr. Silvia Figueira

Position: Professor and Department Chair

Employer: Department of Computer Science and Engineering, *Santa Clara University*

Email: sfigueira@scu.edu

Dr. Wei Wang

Position: Professor (Ph.D. Advisor)

Employer: Department of Computer Science, *San Diego State University*

Email: wwang@sdsu.edu